

Zain Rizvi

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EDUCATION

University of Ottawa

Honours BSc Computer Science, Co-op

Ottawa, ON

Sept. 2023 – Dec. 2027

PROJECTS

NFL Contract Value Analyzer | *Python, XGBoost, scikit-learn, K-Means, pandas, Streamlit, Plotly*

- [nfl-contract-analyzer.streamlit.app](#) | [Demo](#) | [GitHub](#)
- Trained separate XGBoost regression models per position to predict fair market APY for 264 active NFL players, achieving R^2 of 0.96 for QBs and MAE under \$1M for WRs and TEs
- Engineered 40+ features including injury-adjusted stats scaled to a 17-game pace, per-game production rates, career peak stats, and guarantee percentage to normalize across players with varying availability
- Applied K-Means clustering on per-game production rates and age to discover 4 unsupervised archetypes per position, surfacing groupings like WR1/Alpha and Slot Technician without predefined labels
- Built a deterministic data pipeline joining nflverse seasonal stats with Over The Cap contract data on GSIS player ID, processing 50,000+ raw records down to 89 legitimate market deals for model training

GitHub Secret Scanner | *Python, Shannon entropy, regex, PyGithub, Streamlit, Plotly*

- [git-scanner.streamlit.app](#) | [Demo](#) | [GitHub](#)
- Built a two-layer credential detection engine scanning public GitHub repositories using regex pattern matching across 14 secret types and Shannon entropy analysis to flag high-randomness strings that patterns miss
- Each finding includes real-world impact assessment, the exact attack vector an attacker would use, and step-by-step remediation guidance, modeled after enterprise tools like GitGuardian and TruffleHog
- Deployed an interactive dashboard with terminal-style UI, severity breakdowns by detection method, and CSV export for audit reporting

FilmGrid | *React, Node.js, Express, MongoDB, SerpApi, Cheerio, Mapbox, Vercel, Railway*

- [filmgrid-flax.vercel.app](#) | [Demo](#) | [GitHub](#)
- Built a full-stack movie theatre finder surfacing real-time showtimes for 15+ Toronto theatres using the SerpApi Google Maps engine with geolocation-based distance sorting
- Engineered Cheerio web scrapers for indie theatres including Revue Cinema and Paradise Theatre, providing showtime data unavailable on Fandango or Google Movies
- Deployed a decoupled architecture with the React frontend on Vercel and Node/Express backend on Railway, with in-memory caching on a 2-hour TTL and TMDb API integration for movie metadata

EXPERIENCE

IT Service Desk Analyst

May 2024 – Aug. 2024, May 2025 – Aug. 2025

York Region

Newmarket, ON

- Triaged and investigated 500+ monthly IT incidents across 15+ service categories, identifying recurring issue patterns and coordinating with technicians to resolve root causes
- Built dashboards tracking incident volume, priority, and resolution time, surfacing trends used to inform infrastructure and staffing decisions across a team of 10+ technicians
- Standardized ticket taxonomy and documentation practices to improve data consistency and enable accurate security and operational reporting for management
- Coordinated with vendors and internal teams to manage software, hardware, and access-related requests, maintaining accurate records for audit and compliance purposes

Director of Academic Affairs

Sept. 2023 – May 2024

Computer Science Student Association (CSSA)

University of Ottawa

- Organized professional development events including a Nokia office tour for 50+ CS students, handling logistics and outreach end to end
- Served as the primary liaison between the student body and department administration on academic policy and programming

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Java, HTML/CSS, R

Security & ML: Shannon entropy, regex, XGBoost, scikit-learn, pandas, K-Means clustering, Plotly, Streamlit

Web: React, Node.js, Express, Three.js, MongoDB, REST APIs

Tools: Git, AWS, Docker, PostgreSQL, Nginx, VS Code